

21. Let a tangent be drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3} \cos \theta, \sin \theta)$ where $\theta \in \left(0, \frac{\pi}{2}\right)$. Then the value of θ such that the sum of intercepts on axes made by this tangent is minimum is equal to -

माना दीर्घवृत्त $\frac{x^2}{27} + y^2 = 1$ के बिन्दु $(3\sqrt{3} \cos \theta, \sin \theta)$, $\theta \in \left(0, \frac{\pi}{2}\right)$, पर एक स्पर्श रेखा खींची गई है। θ का वह मान, जिसके लिए इस स्पर्श रेखा द्वारा अक्षों पर बनाए गए अंतःखण्डों का योगफल निम्नतम है, बराबर है :

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{8}$

Ans. B

Sol. $\frac{x^2}{27} + y^2 = 1$

$$\frac{2x}{27} + 2y y' = 0$$

$$\frac{x}{27} + y y' = 0 \Rightarrow yy' = \frac{-x}{27}$$

$$y' = \frac{-x}{27y}$$

$$y'_{(3\sqrt{3} \cos \theta, \sin \theta)} = \frac{-3\sqrt{3} \cos \theta}{27 \sin \theta} = \frac{-\sqrt{3} \cos \theta}{9 \sin \theta}$$

$$y' = \frac{-\cot \theta}{3\sqrt{3}}$$

$$y - \sin \theta = \frac{-\cot \theta}{3\sqrt{3}} (x - 3\sqrt{3} \cos \theta)$$

Put $x = 0$

$$y - \sin \theta = \cos \theta \times \cot \theta$$

$$y = \cos \theta \cot \theta + \sin \theta = \frac{1}{\sin \theta}$$

Put $y = 0$

$$-\sin \theta = \frac{-\cot \theta}{3\sqrt{3}} (x - 3\sqrt{3} \cos \theta)$$

$$\sin \theta = \frac{\cos \theta}{3\sqrt{3} \sin \theta} (x - 3\sqrt{3} \cos \theta)$$

$$\frac{3\sqrt{3} \sin^2 \theta}{\cos \theta} = x - 3\sqrt{3} \cos \theta$$

$$\frac{3\sqrt{3} \sin^2 \theta + 3\sqrt{3} \cos^2 \theta}{\cos \theta} = x$$

$$x = \frac{3\sqrt{3}}{\cos \theta}$$

$$\begin{aligned} x_{\text{int}} + y_{\text{int}} &= \frac{1}{\sin \theta} + \frac{3\sqrt{3}}{\cos \theta} \\ &= \operatorname{cosec} \theta + 3\sqrt{3} \sec \theta \end{aligned}$$

$$f(\theta) = 3\sqrt{3} \sec \theta + \operatorname{cosec} \theta$$

$$f'(\theta) = 3\sqrt{3} \times \sec \theta \tan \theta - \operatorname{cosec} \theta \cot \theta$$

$$= \frac{3\sqrt{3}}{\cos \theta} \times \frac{\sin \theta}{\cos \theta} = \frac{\cos \theta}{\sin^2 \theta}$$

$$3\sqrt{3} = \cot^3 \theta$$

$$3^{3/2} = \cot^3 \theta$$

$$\cot \theta = \sqrt{3}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

22. The position of a moving car at time t is given by $f(t) = at^2 + bt + c$, $t > 0$, where a , b and c are real numbers greater than 1. Then the average speed of the car over the time interval $[t_1, t_2]$ is attained at the point :

एक गतिशी कार की t समय पर स्थिति $f(t) = at^2 + bt + c$, $t > 0$ द्वारा दी गई है, जहाँ $a > 1$, $b > 1$ तथा $c > 1$ वास्तविक संख्याएँ हैं, तो समय अंतराल $[t_1, t_2]$ में कार की औसत गति निम्न में से किस बिन्दु पर प्राप्त होती है :

- (A) $2a(t_1 + t_2) + b$ (B) $\frac{(t_2 - t_1)}{2}$ (C) $a(t_2 - t_1) + b$ (D) $\frac{(t_1 + t_2)}{2}$

Ans. D

Sol. $f'(t) = \frac{f(t_2) - f(t_1)}{t_2 - t_1} = V_{\text{avg}} \quad (\text{LMVT})$

$$2at + b = \frac{a(t_2^2 - t_1^2) + b(t_2 - t_1)}{t_2 - t_1}$$

$$t = \frac{t_1 + t_2}{2}$$

23. If $(a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$, where $a > b > 0$, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is :

यदि $(a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$, जहाँ $a > b > 0$ हैं, तो $\frac{dx}{dy}$ पर $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{a-2b}{a+2b}$ (B) $\frac{2a+b}{2a-b}$ (C) $\frac{a-b}{a+b}$ (D) $\frac{a+b}{a-b}$

Ans. D

Sol. $(a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$

Differentiate w.r.t. 'x'

$$(0 - \sqrt{2}b \sin x)(a - \sqrt{2}b \cos y) + (a + \sqrt{2}b \cos x) \cdot (0 + \sqrt{2}b \sin y) y' = 0$$

At $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$

$$(-b)(a-b) + (a+b)by' = 0$$

$$y' = \frac{a-b}{a+b}$$

$$\frac{dy}{dx} = \frac{a-b}{a+b} \quad \boxed{\therefore \frac{dx}{dy} = \frac{a+b}{a-b}}$$

24. If $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3}$ is equal to L, then the value of $(6L + 1)$ is :

यदि $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3} = L$ है, तो $(6L + 1)$ का मान है:

(A) $\frac{1}{2}$ (B) 6 (C) $\frac{1}{6}$ (D) 2

Ans. D

Sol. $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3}$

This is $\frac{0}{0}$ Indeterminate form

Applying L'Hopital Rule

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1-x^2}} - \frac{1}{1+x^2}}{9x^2} \\ &= \lim_{x \rightarrow 0} \frac{(1+x^2 - \sqrt{1-x^2})(1+x^2\sqrt{1-x^2})}{(\sqrt{1+x^2})(1+x^2)9x^2(1+x^2 + \sqrt{1-x^2})} \\ &= \lim_{x \rightarrow 0} \frac{(1+x^2)^2 - (1-x^2)}{9x^2 \times 2} \\ &= \lim_{x \rightarrow 0} \frac{x^4 + 3x^2}{18x^2} = \frac{1}{6} \\ \Rightarrow 6L + 1 &= 2 \end{aligned}$$

25. Let $y = f(x) = \sin^3 \left(\frac{\pi}{3} \left(\cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right) \right)$. Then, at $x = 1$,

माना $y = f(x) = \sin^3 \left(\frac{\pi}{3} \left(\cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right) \right)$ है। तो $x = 1$ पर

(A) $\sqrt{2}y' - 3\pi^2 y = 0$

(B) $y' + 3\pi^2 y = 0$

(C) $2y' + 3\pi^2 y = 0$

(D) $2y' + \sqrt{3}\pi^2 y = 0$

Ans. C

Sol. $Y = f(x) = \sin^3 \left(\frac{\pi}{3} \left(\cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right) \right)$

$$\frac{dy}{dx} = 3 \sin^2 \left(\frac{\pi}{3} \cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right) \times \frac{\pi}{3} \left(-\sin \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{\frac{3}{2}} \right) \right)^{\frac{3}{2}}$$

$$\left(\cos \left(\frac{\pi}{3} \cos \left(\frac{\pi}{3\sqrt{2}} (-4x^3 + 5x^2 + 1)^{3/2} \right) \times \frac{\pi}{3\sqrt{2}} (3/2(-4x^3 + 5x^2 + 1)^{1/2})(-12x^2 + 10x) \right) \right)$$

at $x = 1$

$$y = f(1) = \sin^3 \left(\frac{\pi}{3} \left(\cos \left(\frac{\pi}{3\sqrt{2}} \cdot 2\sqrt{2} \right) \right) \right)$$

$$= \sin^3 \left(\frac{\pi}{3} (-1/2) \right)$$

$$= \sin^3 \left(-\frac{\pi}{6} \right) = \left(-\frac{1}{2} \right)^3 = -\frac{1}{8}$$

$$\left(\frac{dy}{dx} \right)_{x=1} = 3 \left(-\frac{1}{2} \right)^2 \cdot \left(-\frac{\pi}{3} \right) \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) \cdot \frac{\pi}{3\sqrt{2}} \left(\frac{3}{2} \sqrt{2} \right) (-2)$$

$$= \frac{3}{4} \cdot \frac{\pi}{3} \cdot \frac{3}{4} = \frac{3\pi^2}{16}$$

Now (3) $2y' + 3\pi^2 y = 2 \left(\frac{3\pi^2}{16} \right) + 3\pi^2 \left(-\frac{1}{8} \right) = \frac{3\pi^2}{8} - \frac{3\pi^2}{8} = 0$

26. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \left(2 \sin \left(\frac{1}{x} \right) \right) |x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then f is:

- (A) monotonic on $(-\infty, 0)$ only
 (B) monotonic on $(0, \infty)$ only
 (C) not monotonic on $(-\infty, 0)$ and $(0, \infty)$
 (D) monotonic on $(-\infty, 0) \cup (0, \infty)$

$$f(x) = \begin{cases} \left(2 \sin \left(\frac{1}{x} \right) \right) |x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

द्वारा परिभाषित फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ का विचार कीजिए। फलन f

- (A) केवल $(-\infty, 0)$ पर एकदिष्ट है
 (B) केवल $(0, \infty)$ पर एकदिष्ट (monotonic) है

(C) $(-\infty, 0)$ तथा $(0, \infty)$ पर एकदिष्ट नहीं है

(D) $(-\infty, 0) \cup (0, \infty)$ पर एकदिष्ट है

Ans. C

Sol.
$$f(x) = \begin{cases} \left(2 - \sin \frac{1}{x}\right)x & ; x > 0 \\ -\left(2 - \sin \frac{1}{x}\right)x & ; x < 0 \\ 0 & ; x = 0 \end{cases}$$

$$f'(x) = \begin{cases} \left(2 - \sin x \frac{1}{x}\right) + \frac{1}{x} \cos \frac{1}{x^2} & ; x > 0 \\ -\left(2 + \sin \frac{1}{x}\right) - \frac{1}{x} \cos \frac{1}{x^2} & ; x < 0 \\ 0 & ; x = 0 \end{cases}$$

$$f'(x) = \begin{cases} 2 - \sin x \frac{1}{x} + \frac{1}{x} \cos \frac{1}{x^2} & ; x > 0 \\ -2 + \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x^2} & ; x < 0 \\ 0 & ; x = 0 \end{cases}$$

Clearly here $f'(x)$ is a oscillating function which is not monotonic in $(-\infty, 0) \cup (0, \infty)$.

27. If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$ then $\frac{dy}{dx}$ is equal to

यदि $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$ है, तो $\frac{dy}{dx}$ बराबर है :

- (A) $2x - \frac{\pi}{3}$ (B) $x - \frac{\pi}{6}$ (C) $\frac{\pi}{3} - x$ (D) $\frac{\pi}{6} - x$

Ans. B

sol. $2y = \left[\cot^{-1} \left(\frac{\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x}{\frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x} \right) \right]^2$

$$\Rightarrow 2y = 2y = \left[\cot^{-1} \left(\frac{\cos\left(\frac{\pi}{6} - x\right)}{\sin\left(\frac{\pi}{6} - x\right)} \right) \right]^2$$

$$\Rightarrow 2y = \left[\cot^{-1} \left(\cot\left(\frac{\pi}{6} - x\right) \right) \right]^2 \quad \because \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, \frac{\pi}{6} \right)$$

$$\Rightarrow 2y = \begin{cases} \left(\frac{7\pi}{6} - x \right)^2 & \text{if } \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, 0 \right) \\ \left(\frac{\pi}{6} - x \right)^2 & \text{if } \frac{\pi}{6} - x \in \left(0, \frac{\pi}{6} \right) \end{cases}$$

$$\frac{dy}{dx} = \begin{cases} x - \frac{7\pi}{6} & \text{if } x \in \left(\frac{\pi}{6}, \frac{\pi}{2} \right) \\ x - \frac{\pi}{6} & \text{if } x \in \left(0, \frac{\pi}{6} \right) \end{cases}$$

Note: only one given option is correct

28. If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$; then the set

$S = \{x \in \mathbb{R} : f(x) = f(0)\}$ contains exactly :

(A) Four irrational numbers

(B) Four rational numbers

(C) Two irrational and one rational number

(D) Two irrational and two rational numbers

यदि $f(x)$, घात चार का एक शून्येत्तर बहुपद है, जिसके स्थानीय चरम बिन्दु $x = -1, 0, 1$ पर हैं, तो समुच्चय

$S = \{x \in \mathbb{R} : f(x) = f(0)\}$ रखता है -

(A) चार अपरिमेय संख्याएँ

(B) चार परिमेय संख्याएँ

(C) दो अपरिमेय तथा एक परिमेय संख्या

(D) दो अपरिमेय तथा दो परिमेय संख्याएँ

Ans. C

sol. $f'(x) = A(x+1)x(x-1)$

$$= A(x^3 - x)$$

$$\Rightarrow f(x) = A\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + C$$

Now $f(0) = C$

$$\therefore f(x) = f(0) \Rightarrow A\left(\frac{x^4}{4} - \frac{x^2}{2}\right) = 0$$

$$\Rightarrow \frac{x^2}{2}\left(\frac{x^2}{2} - 1\right) = 0$$

$$\Rightarrow x = 0, -\sqrt{2}, \sqrt{2}$$

$$\therefore S = \{0, -\sqrt{2}, \sqrt{2}\}$$

29. The function $f(x) = \frac{4x^3 - 3x^2}{6} - 2\sin x + (2x-1)\cos x$:

(A) Decreases in $\left[\frac{1}{2}, \infty\right)$

(B) Decreases in $\left(-\infty, \frac{1}{2}\right]$

(C) Increases in $\left(-\infty, \frac{1}{2}\right]$

(D) Increases in $\left[\frac{1}{2}, \infty\right)$

फलन $f(x) = \frac{4x^3 - 3x^2}{6} - 2\sin x + (2x-1)\cos x$:

(A) $\left[\frac{1}{2}, \infty\right)$ में ह्रासमान है

(B) $\left(-\infty, \frac{1}{2}\right]$ में ह्रासमान है

(C) $\left(-\infty, \frac{1}{2}\right]$ में वर्धमान है

(D) $\left[\frac{1}{2}, \infty\right)$ में वर्धमान है

Ans. D

Sol. $f'(x) = (2x-1)(x-\sin x) \geq 0$ for $x \in \left[\frac{1}{2}, \infty\right)$

$f(x)$ is increasing for $x \in \left[\frac{1}{2}, \infty\right)$

30. Let f be a twice differentiable function on $(1, 6)$. If $f(2) = 8$, $f'(2) = 5$, $f'(x) \geq 1$ and $f''(x) \geq 4$, for all $x \in (1, 6)$, then :

माना f , अन्तराल $(1, 6)$ में दो बराबर अवकलनीय है। यदि $f(2) = 8$, $f'(2) = 5$, तथा सभी $x \in (1, 6)$ के लिए $f'(x) \geq 1$ तथा $f''(x) \geq 4$ हैं, तो :

- (A) $f(5) + f'(5) \geq 28$ (B) $f(5) + f'(5) \leq 26$
(C) $f(5) \leq 10$ (D) $f'(5) + f''(5) \leq 20$

Ans. A

Sol. We have

$$f'(x) \geq 1$$

$$\int_2^5 f'(x) dx \geq \int_2^5 1 dx$$

$$\Rightarrow [f(x)]_2^5 \geq (5 - 2)$$

$$\Rightarrow f(5) - f(2) \geq 3$$

$$\Rightarrow f(5) \geq 3 + f(2)$$

$$f(5) \geq 11 \quad \dots\dots\dots(I)$$

$$f''(x) \geq 4$$

$$\int_2^5 f''(x) dx \geq \int_2^5 4 dx$$

$$\Rightarrow [f'(x)]_2^5 \geq 4(5 - 2)$$

$$\Rightarrow f'(5) - f'(2) \geq 12$$

$$\Rightarrow f'(5) \geq 12 + f'(2) = 17$$

$$\therefore f(5) \geq 17 \quad \dots\dots\dots(II)$$

using (I) and (II)

$$f(5) + f'(5) \geq 28$$

31. Let $f(x) = (\sin(\tan^{-1} x) + \sin(\cot^{-1} x))^2 - 1$, $|x| > 1$. If $\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\sin^{-1}(f(x)))$ and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

माना $f(x) = (\sin(\tan^{-1} x) + \sin(\cot^{-1} x))^2 - 1$, $|x| > 1$ है। यदि $\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\sin^{-1}(f(x)))$ तथा $y(\sqrt{3}) = \frac{\pi}{6}$ हैं, तो

$y(-\sqrt{3})$ का मान है :

- (A) $\frac{5\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $-\frac{\pi}{6}$

Ans. A

Sol. $f(x) = \left(\sin(\tan^{-1} x) + \sin\left(\frac{\pi}{2} - \tan^{-1} x\right) \right)^2 - 1$

$$f(x) = 1 + 2 \sin(\tan^{-1} x) \cos(\tan^{-1} x) - 1$$

$$f(x) = \sin(2 \tan^{-1} x)$$

$$\sin^{-1} f(x) = \sin^{-1} \sin(2 \tan^{-1} x)$$

$$\sin^{-1} f(x) = \pi - 2 \tan^{-1} x \quad \because x = \sqrt{3}$$

$$\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\pi - 2 \tan^{-1} x)$$

$$y = -\tan^{-1} x + c$$

$$\frac{\pi}{6} = -\frac{\pi}{3} + c$$

$$c = \frac{\pi}{2}$$

$$y = -\tan^{-1} x + \frac{\pi}{2}$$

$$y = \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$$

32. The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point :

वक्र $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ की अधिकतम प्रवणता निम्न में से किस बिंदु पर है ?

- (A) (0, 0) (B) $\left(3, \frac{21}{2}\right)$ (C) (2, 2) (D) (2, 9)

Ans. C

Sol. Slope $= \frac{dy}{dx} = 2x^3 - 15x^2 + 36x - 19 = m$

To find maximum : differentiating

$$\frac{dm}{dx} = 6x^2 - 30x + 36 = 0 \Rightarrow x = 2 \text{ OR } 3$$

$$\begin{array}{c} + \quad \cap \quad - \quad \cup \quad + \\ | \quad \quad | \\ 2 \quad \quad 3 \\ \text{max} \quad \text{min} \end{array}$$

$x = 2$; point of maxima

$x = 3$; point of minima

So point of maximum ; $x = 2$;

$$\begin{aligned} y &= \frac{1}{2}(2)^4 - 5(2)^3 + 18(2)^2 - 19(2) \\ &= 8 - 40 + 72 - 38 \\ &= 2 \\ &= (2, 2) \end{aligned}$$

33. The triangle of maximum area that can be inscribed in a given circle of radius 'r' is :

(A) An equilateral triangle of height $\frac{2r}{3}$.

(B) A right angle triangle having two of its sides of length $2r$ and r .

(C) An equilateral triangle having each of its side of length $\sqrt{3} r$.

(D) An isosceles triangle with base equal to $2r$.

'r' त्रिज्या के एक वृत्त के अंतर्गत अधिकतम क्षेत्रफल का त्रिभुज निम्न में से कौन सा है?

(A) एक समबाहु त्रिभुज जिसकी ऊँचाई $\frac{2r}{3}$ है

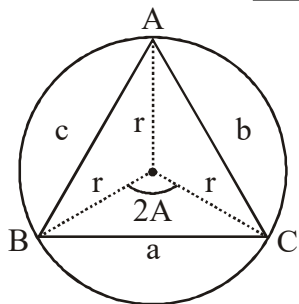
(B) एक समकोण त्रिभुज जिसकी दो भुजाओं की लम्बाई $2r$ तथा r है

(C) एक समबाहु त्रिभुज जिसकी प्रत्येक भुजा की लम्बाई $\sqrt{3} r$ है।

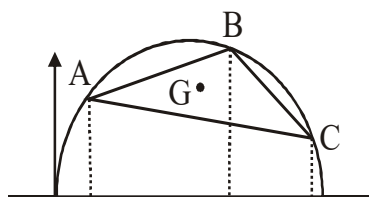
(D) एक समद्विबाहु त्रिभुज जिसका आधार $2r$ है

Ans. C

Sol. Area of $\triangle ABC$



$$\Delta = \frac{abc}{4r}$$



$$\text{Centroid } G = \left(\frac{A + B + C}{3}, \frac{\sin A + \sin B + \sin C}{3} \right)$$

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin \left(\frac{A + B + C}{3} \right)$$

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

$$\frac{a}{2r} + \frac{b}{2r} + \frac{c}{2r} \leq \frac{3\sqrt{3}}{2}$$

$$\Rightarrow \frac{a + b + c}{3} \leq \sqrt{3} r$$

$$GM \leq AM$$

$$(abc)^{\frac{1}{3}} \leq \frac{a + b + c}{3}$$

$$\Rightarrow abc \leq \left(\frac{a + b + c}{3} \right)^3$$

$$\Rightarrow \frac{abc}{4r} \leq \frac{1}{4r} \left(\frac{a + b + c}{3} \right)^3$$

$$\Rightarrow \Delta \leq \frac{1}{4r} \left(\frac{a + b + c}{3} \right)^3 \leq \frac{1}{4r} (\sqrt{3} r)^3$$

$$\Delta \leq \frac{3\sqrt{3}}{4} r^2$$

$$\Delta_{\max} = \frac{3\sqrt{3}}{4} r^2$$

(Equality holds when $a = b = c$)

Δ is an equilateral Δ

$$a = \sqrt{3} r$$

34. Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}, x \in \mathbb{R}$, where a, b and d are non-zero real constants. Then :

(A) f is neither increasing nor decreasing function of x

(B) f' is not a continuous function of x

(C) f is a decreasing function of x

(D) f is an increasing function of x

माना $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}, x \in \mathbb{R}$ जहाँ a, b तथा d शून्येतर वास्तविक अचर हैं, तो –

(A) f, x का न तो वर्धमान न ही ह्यसमान फलन है।

(B) f', x का संतत फलन नहीं है।

(C) f, x का ह्यसमान फलन है।

(D) f, x का एक वर्धमान फलन है।

Ans. D

Sol.
$$f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$$

$$f'(x) = \frac{a^2}{(a^2 + x^2)^{3/2}} + \frac{b^2}{(b^2 + (d-x)^2)^{3/2}} > 0 \forall x \in \mathbb{R}$$

$\therefore f(x)$ is an increasing function

35. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water

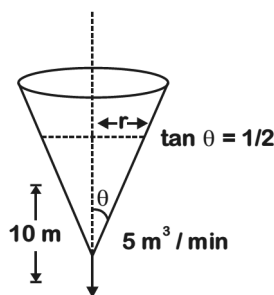
is poured into it at a constant rate of 5 cubic metre per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10 m; is :

एक पानी की टंकी उल्टे लंब वृत्तीय शंकु के आकार की है, जिसका अर्ध-शीर्ष कोण $\tan^{-1}\left(\frac{1}{2}\right)$ है। इसमें पानी 5 घन मीटर प्रति मिनट की समान दर से डाला जाता है, तो टंकी में पानी की गहराई 10 m होने पर वह दर (मी./मि. में), जिस पर पानी की सतह बढ़ रही है, है:

- (A) $\frac{2}{\pi}$ (B) $\frac{1}{15\pi}$ (C) $\frac{1}{10\pi}$ (D) $\frac{1}{5\pi}$

Ans. D

sol.



Given $\frac{dv}{dt} = 5 \text{ m}^3 / \text{min}$

$V = \frac{1}{3} \pi r^2 h$ (i) (where r is radius and h is height at any time)

Also $\tan \theta = \frac{r}{h} = \frac{1}{2} \Rightarrow h = 2r \Rightarrow \frac{dh}{dt} = \frac{2dr}{dt}$ (ii)

Differentiate eqⁿ. (i), we get

$$\frac{dV}{dt} = \frac{1}{3} \left(\pi 2r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt} \right) = \frac{1}{3} \left(100\pi \frac{1}{2} + 25\pi \right) \frac{dh}{dt}$$

at $h = 10, r = 5$

$$5 = \frac{75\pi}{3} \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{1}{5\pi} \text{ m / min}$$

36. Let $f : (-1, \infty) \rightarrow \mathbb{R}$ be defined by $f(0) = 1$ and $f(x) = \frac{1}{x} \log_e (1 + x)$, $x \neq 0$. Then the function f :

- (A) decreases in $(-1, \infty)$. (B) decreases in $(-1, 0)$ and increases in $(0, \infty)$.
(C) increases in $(-1, \infty)$. (D) increases in $(-1, 0)$ and decreases in $(0, \infty)$.

यदि $f: (-1, \infty) \rightarrow \mathbb{R}$, $f(0) = 1$ तथा $f(x) = \frac{1}{x} \log_e (1+x)$, $x \neq 0$ द्वारा परिभाषित है, तो फलन f :

(A) $(-1, \infty)$ में ह्रासमान है।

(B) $(-1, 0)$ में ह्रासमान है तथा $(0, \infty)$ में वर्धमान है।

(C) $(-1, \infty)$ में वर्धमान है।

(D) $(-1, 0)$ में वर्धमान है तथा $(0, \infty)$ में ह्रासमान है।

Ans. A

Sol.
$$f'(x) = \frac{x \cdot \frac{1}{1+x} - \ln(1+x) \cdot 1}{x^2}$$

$$f'(x) = \frac{x - (1+x) \ln(1+x)}{x^2(1+x)}$$

Let $h(x) = x - (1+x) \ln(1+x)$

$$h'(x) = 1 - (\ln(1+x) + 1)$$

$$\Rightarrow h'(x) = -\ln(1+x)$$

Sign of $h'(x)$: $\begin{array}{c} | \quad + \quad | \quad - \\ -1 \quad \quad 0 \end{array}$

$g(x)$ is maximum at $x = 0$

$$g(x)|_{\max} = g(0) = 0$$

$$g(x) < 0$$

$$f'(x) < 0$$

$f(x)$ is always decreasing in $(-1, \infty)$.

37. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$

Let $A = \{x \in \mathbb{R} : f \text{ is increasing}\}$. Then A is equal to :

माना $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} -55x, & \text{यदि } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{यदि } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{यदि } x > 4, \end{cases}$$

द्वारा परिभाषित है। माना $A = \{x \in \mathbb{R} : f \text{ वर्धमान है}\}$ । तो A बराबर है :

- (A) $(-5, \infty)$ (B) $(-\infty, -5) \cup (4, \infty)$
 (C) $(-5, -4) \cup (4, \infty)$ (D) $(-\infty, -5) \cup (-4, \infty)$

Ans. C

Sol.

$$f(x) = \begin{cases} -55x & x < -5 \\ 2x^3 - 3x^2 - 120x & -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336 & x > 4 \end{cases}$$

$$f'(x) = \begin{cases} -55 & x < -5 \\ 6x^2 - 6x - 120 & -5 < x < 4 \\ 6x^2 - 6x - 36 & x > 4 \end{cases}$$

C - I $x < -5$

$f'(x) < 0$

$\Rightarrow f$ is decreasing

C - II $-5 < x < 4$

$f'(x) > 0 \Rightarrow x \in (-\infty, -4) \cup (5, \infty)$

$\Rightarrow x \in (-5, -4) \Rightarrow f$ is increasing

C - III $x > 4$

$f'(x) > 0$

$6(x^2 - x - 6) > 0 \Rightarrow x \in (-\infty, -2) \cup (3, \infty)$

$\Rightarrow x \in (4, \infty) \Rightarrow f$ is increasing

38. For all twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, with $f(0) = f(1) = f'(0) = 0$,

- (A) $f'(x) = 0$, for some $x \in (0, 1)$ (B) $f'(x) \neq 0$ at every point $x \in (0, 1)$
 (C) $f''(0) = 0$ (D) $f'(x) = 0$, at every point $x \in (0, 1)$

प्रत्येक दो बार अवकलनीय फलन $f : \mathbb{R} \rightarrow \mathbb{R}$ जिसके लिए $f(0) = f(1) = f'(0) = 0$ है, तो :

(A) किसी $x \in (0,1)$ के लिए $f'(x) = 0$

(B) प्रत्येक बिन्दु $x \in (0,1)$ पर $f'(x) \neq 0$

(C) $f'(0) = 0$

(D) प्रत्येक बिन्दु $x \in (0,1)$ पर $f'(x) = 0$

Ans. A

Sol. $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(0) = f(1) = f'(0) = 0$

Apply Rolle's theorem on $y = f(x)$ in $x \in [0, 1]$

$$f(0) = f(1) = 0$$

$$\Rightarrow f'(\alpha) = 0 \text{ where } \alpha \in (0,1)$$

Now apply Rolle's theorem on $y = f'(x)$

in $x \in [0, \alpha]$

$$f'(0) = f'(\alpha) = 0 \text{ (} f'(x) \text{ is continuous and differentiable)}$$

$$\Rightarrow f''(\beta) = 0 \text{ for some } \beta \in (0, \alpha)$$

$$\Rightarrow f''(x) = 0 \text{ for some } x \in (0,1)$$

INTEGER TYPE QUESTIONS

39. The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = \alpha$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is _____.

α का न्यूनतम मान, जिसके लिए समीकरण $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = \alpha$ का अन्तराल $\left(0, \frac{\pi}{2}\right)$ में कम से कम एक हल है, है _____।

Ans. 9

Sol. $f(x) = \frac{4}{\sin x} + \frac{1}{1 - \sin x} = \frac{4 - 3 \sin x}{\sin x(1 - \sin x)}$

$$f'(x) = \frac{(2 - \sin x)(3 \sin x - 2)}{\sin^2 x(1 - \sin x)^2}$$

$$f'(x) \Rightarrow \frac{(-)ve \quad 0 \quad (+)ve}{\sin x = 2/3}$$

$$\alpha_{\min} = \frac{4}{2/3} + \frac{1}{1 - 2/3} = 9$$

40. If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola, $y^2 = 4x$ at the point $(1, 2)$ intersect at the same point on the x -axis, then the value of c is _____.

यदि वक्र $y = e^x$ के बिन्दु (c, e^c) पर स्पर्श रेखा तथा परवलय $y^2 = 4x$ के बिन्दु $(1, 2)$ पर अभिलम्ब दोनों x -अक्ष के एक

ही बिन्दु से होकर जाते हैं, तो c का मान है _____।

Ans. 4

Sol. $y^2 = 4x$

$$2y \frac{dy}{dx} = 4$$

$$\frac{dy}{dx} = \frac{4}{2 \times 2}$$

$$m_T = 1$$

$$m_N = -1$$



$$y - 2 = m_N (x - 1)$$

$$y - 2 = -1 (x - 1)$$

$$x + y - 3 = 0 \quad \dots(i)$$

$$y = e^x \quad (c, e^c)$$

$$\frac{dy}{dx} = e^x$$

$$\left. \frac{dy}{dx} \right|_{(c, e^c)} = e^c$$

$$y - e^c = e^c (x - c) \quad \dots(ii)$$

$$y = 0$$

$$x = 3$$

$$x = -1 + c$$

$$c - 1 = 3$$

$$c = 4$$